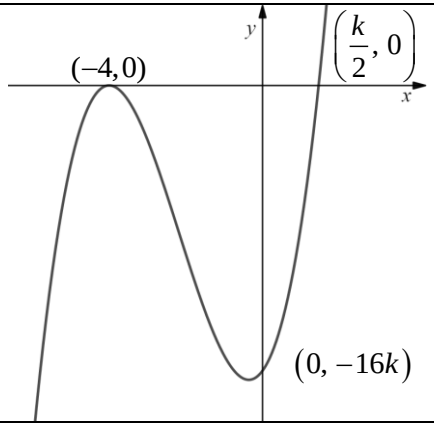
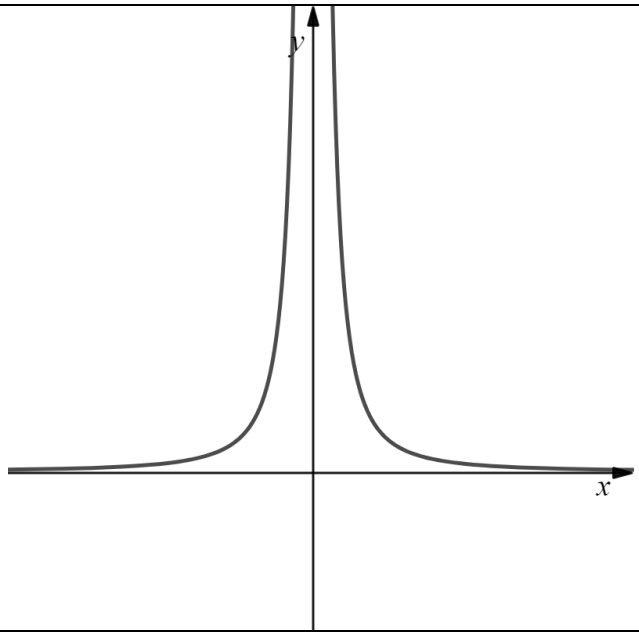
1(a)	$\left(\frac{dy}{dx}=\right) \ldots x^2+\ldots x+\ldots x^{-2}$	M1
	$\left(\frac{dy}{dx}=\right) \frac{3}{4}x^2-2x-17x^{-2}$	A1A1
		(3)
(b)	$\left(\frac{dy}{dx}=\right)\frac{3}{4}(2)^2-2(2)-17(2)^{-2}=-\frac{21}{4}$	M1
	$y-\frac{13}{2}="-\frac{21}{4}"(x-2)$	dM1
	$21x+4y-68=0$	A1
		(3)
		(6 marks)

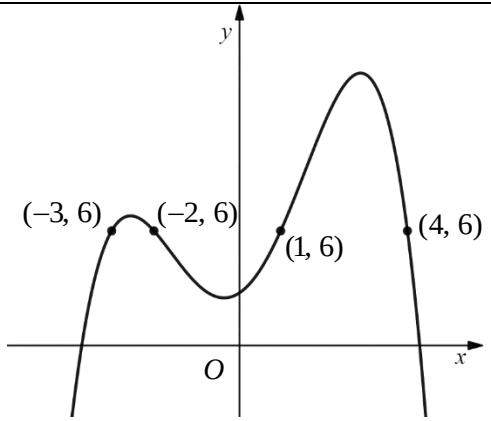
2(a)	$a = 2$	B1
	$b = -3$	B1
		(2)
(b)	Any two term of $\int \frac{2x^3 - 3x^2 - 32x - 15}{5\sqrt{x}}\,\mathrm{d}x = \int \frac{2}{5}x^{\frac{5}{2}} - \frac{3}{5}x^{\frac{3}{2}} - \frac{32}{5}x^{\frac{1}{2}} - 3x^{-\frac{1}{2}}\,\mathrm{d}x$	M1A1
	$x^n \rightarrow x^{n+1}$	M1
	$\frac{4}{35}x^{\frac{7}{2}} - \frac{6}{25}x^{\frac{5}{2}} - \frac{64}{15}x^{\frac{3}{2}} - 6x^{\frac{1}{2}} + c$	A1A1
		(5)
		(7 marks)

3(a)	$1.05 = 3p + q$ $1.65 = 5p + q$ e.g. $\Rightarrow 2p = 0.6$ or $\Rightarrow 1.05 = 3p + (1.65 - 5p)$ or $\Rightarrow 1.65 = 5p + (1.05 - 3p)$ $p = 0.3 \quad q = 0.15$	M1 A1A1
		(3)
(b)	$2.5 = "0.3"T + "0.15" \Rightarrow T =$ $T = 7.8$	M1 A1
		(2)
		(5 marks)

4(a)	$x^2(2x+1)-15x \Rightarrow 2x^3+x^2-15x = x(2x^2+x-15)$	M1
	$x(2x-5)(x+3)=0 \Rightarrow x=...$	dM1
	Two of $x=0, \frac{5}{2}, -3$	B1
	$x=0, \frac{5}{2}, -3$	A1
		(4)
(b)	$y^{\frac{2}{3}} = \frac{5}{2} \Rightarrow y = \left(\frac{5}{2}\right)^{\frac{3}{2}}$	M1
	$\frac{5}{4}\sqrt{10}$	A1cso
		(2)
		(6 marks)

5(a)	$f'(x) = 12x^{-\frac{1}{2}} + \frac{x}{3} - 4$	
	One of $x^{-\frac{1}{2}} \to x^{\frac{1}{2}}$, $-4 \to -4x$, $x \to x^2$	M1
	$f(x) = \int 12x^{-\frac{1}{2}} + \frac{x}{3} - 4\,dx = 24x^{\frac{1}{2}} + \frac{x^2}{6} - 4x + c$	A1A1
	$8 = 24(9)^{\frac{1}{2}} + \frac{(9)^2}{6} - 4(9) + c \Rightarrow c = ...$	dM1
	$\left(f(x) =\right)\ \ 24x^{\frac{1}{2}} + \frac{x^2}{6} - 4x - \frac{83}{2}$	A1
		(5)
(b)	$f'(9) = \frac{12}{\sqrt{9}} + \frac{9}{3} - 4 \quad (=3)$	M1
	$3 \to -\frac{1}{3}$	dM1
	$y - 8 = "-\frac{1}{3}"(0 - 9)$	M1
	$(0, 11)$	A1
		(4)
		(9 marks)

6(a)(i)		B1B1B1
		(3)
(a)(ii)		B1B1
		(2)
(b)	One root because the two graphs intersect each other once	B1
		(1)
		(6 marks)

7(a)	$-1 < x < 2$ $x < -4,\ x > 3$	M1A1 B1
		(3)
(b)	$(x =)$ 1.5	B1
		(1)
(c)(i)		B1B1B1
(ii)	$-3 \le x \le -2$	B1
		(4)
		(8 marks)

8(a)(i)	$2\pi - \frac{2\pi}{3} = \frac{4\pi}{3}$	B1
	Area of sector $= \frac{1}{2} \times 3^2 \times "\frac{4}{3}\pi" = 6\pi$ (m^2)	M1A1
(ii)	Length of arc $= 3 \times "\frac{4}{3}\pi" \Rightarrow$ Perimeter $= 4\pi + 6$ (m)	M1A1
		(5)
(b)	$\frac{1}{2} \times 3^2 \times \sin\left(\frac{2}{3}\pi\right) = \frac{9\sqrt{3}}{4}$ (m^2)	M1A1
		(2)
(c)	Eg $AB^2 = 3^2 + 3^2 - 2 \times 3 \times 3 \times \cos\left(\frac{2}{3}\pi\right) \Rightarrow AB^2 = 27 \Rightarrow AB = 3\sqrt{3}$ (m) *	M1A1*
		(2)
(d)	$\frac{\sin BAC}{8} = \frac{\sin\left(\frac{\pi}{6}\right)}{3\sqrt{3}} \Rightarrow \sin BAC = ...$ or $BAC = ...$	M1
	$\sin BAC =$ awrt $\frac{4\sqrt{3}}{9}$ or $BAC =$ awrt 0.88 $(0.8785\ldots)$	A1
	Area $ABC = \frac{1}{2} \times 3\sqrt{3} \times 8 \times \sin(\pi - \frac{\pi}{6} - "0.88")$ $(= 20.4896....)$	M1
	Total area = "18.8"+"3.90"+"20.5"	dM1
	= awrt 43 (m^2)	A1
		(5)
		(14 marks)

9(a)	$\frac{1}{2}x^2-10x+22=\frac{1}{2}(x\pm...)^2\pm...$ or states $a=\frac{1}{2}$	B1
	$\frac{1}{2}x^2-10x+22=\frac{1}{2}(x\pm10)^2\pm...$ or states $a=\frac{1}{2}$ and $b=\pm10$	M1
	$\frac{1}{2}x^2-10x+22=\frac{1}{2}(x-10)^2-28$	A1
		(3)
(b)	$("10", "-28")$	B1ftB1ft
		(2)
(c)(i)	Gradient of tangent $=8$	B1
	$\frac{dy}{dx}=x-10=8\Rightarrow x=...$	M1
	$x=18,\ y=4$	A1A1
(c)(ii)	$k-\frac{1}{8}\times"18"="4"\Rightarrow k=\frac{25}{4}$	dM1A1
		(6)
(d)	One of x …"10" or y „ $"\frac{25}{4}"-\frac{1}{8}x$ or y …$\frac{1}{2}x^2-10x+22$	B1ft
	Two of x …"10" or y „ $"\frac{25}{4}"-\frac{1}{8}x$ or y …$\frac{1}{2}x^2-10x+22$	B1ft
	All three of x …10, y „ $\frac{25}{4}-\frac{1}{8}x$ and y …$\frac{1}{2}x^2-10x+22$	B1
		(3)
		(14 marks)